Sushanth Buddhala

202-510-7617 | sushanthbuds@gmail.com | linkedin.com/in/sushanth-buddhala-3357a5267 | github.com/sushanthbuds | sushanthbuds.github.io/portfolio
Fairfax, Virginia

PROFILE

Data Analytics Engineering graduate student with hands-on experience building SQL-driven analytics platforms, ETL pipelines, machine learning models, and interactive dashboards. Proven ability to transform large, messy datasets into actionable insights using Python, SQL, and cloud tools. Strong foundation in statistical modeling, predictive analytics, and data engineering, with a track record of delivering explainable, scalable analytics solutions.

EDUCATION

George Mason University — Virginia, USA
M.S. in Data Analytics Engineering

- ***Relevant Coursework:*** *Analytics & Big Data, Intro to Analytics & Modeling, Principles of Data Management & Mining, Visualization for Analytics, Big Data Essentials, Data Mining for Business Analytics, Interpretable Machine Learning, Data Analytics Project*

Gokaraju Rangaraju Institute of Engineering and Technology — Telangana, India
B.S. in Computer Science Engineering

- ***Relevant Coursework:*** *Linear Algebra Differential Calculus, Programming for Problem Solving, Differential Equation and Vector Calculus, Database Management System, Python Programming, Artificial Intelligence, Data Analytics with Python, Machine Learning, Big Data Analytics, Cloud Computing, Data Warehousing and Data Mining*

TECHNICAL SKILLS

Languages & Tools: Python, SQL, R, DAX, FastAPI, Microsoft Excel, Databricks, Snowflake
Data & ML: Data Analysis, Data Modeling, Statistical Analysis, Data Cleaning, Data Storytelling, ETL Pipelines, Machine Learning, Unsupervised Learning, Relational Databases
Visualization & Cloud: Power BI, Tableau, AWS
AI tools & NLP: RAG Pipelines, OpenAI Embeddings, Cursor.ai, Claude code, ChatGPT

PROJECT EXPERIENCE

Housing Migration Analytics Platform (SQL + RAG) — *Nov 2025 – Jan 2026*

- Designed and implemented a normalized SQLite analytics database containing 395K+ U.S. housing, migration, AHS, and CHAS records, leveraging Cursor AI to accelerate development and optimize indexed tables for analytical querying.
- Built a schema-aware, schema-sensitive RAG pipeline using OpenAI embeddings to generate safe, explainable SQL, preventing hallucinations, destructive queries, and invalid SQL.
- Improved query accuracy and reliability by enforcing schema validation and SQL guardrails, enabling non-technical users to safely explore data.

Predictive Pavement Condition Analysis (ML & Big Data) — *Jan 2025 – May 2025*

- Processed and analyzed 2,000+ pavement condition records using imputation, normalization, encoding, and outlier handling to improve data quality.
- Applied K-Means clustering to identify three pavement deterioration patterns, revealing faster degradation in high-traffic environments.
- Built and evaluated regression models (Gradient Boosting, Ridge, Lasso), achieving best performance with Gradient Boosting (RMSE $\approx$ 890).

Obesity Level Analysis (Statistical & Machine Learning Study) — *Aug 2024 – Nov 2024*

- Conducted exploratory data analysis on 2,000+ records to uncover relationships between BMI, lifestyle factors, and family history.
- Developed and validated Logistic Regression, Decision Tree, SVM, and Clustering models, achieving up to 85% accuracy (best with SVM).
- Identified diet, physical activity, and family history as key predictors, translating findings into actionable public-health insights.

PROFESSIONAL CERTIFICATES

• IBM Data Analyst Professional Certificate by Coursera • AWS Cloud Practitioner • Generative AI for Data Analytics by Coursera